Mojtaba Jafarian Abyaneh

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Summary

PhD candidate (FAU) developing AI-driven conflict detection and trajectory prediction from real-world LiDAR data (NSF/NIH-funded); deep learning, GIS, and federal dataset management for ITS safety, connected vehicles, and traffic operations.

Education

Florida Atlantic University Boca Raton, FL
Ph.D. in Transportation and Environmental Engineering | GPA: 4.0/4.0 *2024 – Present*

- *Focus: AI-driven pedestrian–vehicle conflict detection, trajectory prediction, and ITS safety applications*

University of New Brunswick Fredericton, NB, Canada
M.Sc. in Civil Engineering *2018 – 2020*

Experience

Doctoral Research Assistant Aug. 2024 – Present
Florida Atlantic University *Boca Raton, FL*

- Developed AI models for **pedestrian–vehicle conflict prediction** from **real-world LiDAR data** (12,390+ trajectories, West Palm Beach intersection); published in *IEEE OJ-ITS* (IF: 6.007).
- Developed SIPT: multi-agent trajectory prediction across vehicles, pedestrians, and cyclists; applicable to **V2X safety** and **connected vehicle signal optimization**; under review at *IEEE T-ITS*.
- Managed NIH-funded (R01AG068472) longitudinal dataset: multi-sensor in-vehicle data from 750 older drivers, 12 collection points over 3 years; owned data QA and **federal protocol compliance**.
- Running OpenFOAM CFD simulations on HPC for hurricane wind-load analysis of Miami downtown, assessing pedestrian comfort and resilience.
- Building AI-driven smart sensor analytics for coastal transportation monitoring (NSF CS3, VIP Program); mentoring 2 REU students in GPS-based driver behavior analysis and 3D city reconstruction (**QuASAR**).

Technical Skills

- **Programming & Analysis:** Python, MATLAB, Git, LaTeX, Tableau
- **AI & Data:** Deep Learning (PyTorch, TensorFlow), Computer Vision, LiDAR Processing, Spatial GNNs, Pandas, NumPy, Statistical Analysis
- **GIS & Spatial:** ArcGIS, Geopandas, Spatial Data Analysis **| Engineering:** AutoCAD, MicroStation, Civil 3D, Synchro, OpenFOAM (HPC)

Publications

1. **Jafarian Abyaneh, M.** and Jang, J. (2026) "Transformer-Based Trajectory Prediction Using LiDAR Data for Situational Awareness in Complex Urban Environments." *IEEE Open Journal of Intelligent Transportation Systems*, Vol. 7, pp. 16–28. doi:10.1109/OJITS.2025.3640002 (IF: 6.007)
2. **Jafarian Abyaneh, M.**, Jang, J., Smyth, A. W., and Turkcan, M. K. (2026) "Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction in Urban Streetscapes." *IEEE Transactions on Intelligent Transportation Systems*. [Under Review]
3. **Jafarian Abyaneh, M.**, Jang, J., and Kaisar, E. I. (2025) "Spatial GCN for Predicting Bike-Sharing Origin–Destination Spatiotemporal Flows." *SSRN Working Paper* No. 5801889. [Preprint]

Selected Presentations

- **TRB 105th Annual Meeting**, Washington, DC (Jan. 2026).
- CS3 Innovation Summit Perfect Pitch Competition, Columbia University, New York, NY (Feb. 2026).

Research Funding & Awards

- **NSF** Engineering Research Center for Smart Streetscapes (CS3), Award No. EEC-2133516 — Graduate Researcher
- **NIH** In-Vehicle Sensors to Detect Cognitive Change in Older Drivers, Award No. R01AG068472 — Graduate Researcher
- **FAU VIP** Smart Sensors and AI for Coastal Destination Resilience — Graduate Research Mentor
- **ASCE** Palm Beach Branch Scholarship, $1,000 (2026)